Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*	(i)		[Minimum] energy needed to eject an electron [from a material **or** surface **or** metal **not atom**]	1			1		
		(ii)	I	Brighter light of same frequency means more photons [per second] but individual photons unchanged (accept hf unchanged) (1) [so] ejection rate increased (1) accept more electrons emitted but maximum KE unchanged (1)	3			3		
			II	Photon energy for frequency 7.4×10^{14} Hz is 4.9×10^{-19} J or by implication (1) $\phi = 3.7 \times 10^{-19}$ [J] (1) Photon energy for frequency 5.1×10^{14} Hz is 3.4×10^{-19} [J] **or** threshold frequency = 5.59×10^{14} Hz (1) or by implication **ecf** on arithmetic slip on ϕ Photon energy < work function **or** photon frequency < threshold frequency s<u>o no emission </u>or equivalent (1) no **ecf** Statement that KE is negative treat as neutral			4	4	3	
	(b)	(i)		Photon energy expressed as $E = \frac{hc}{\lambda}$ or by implication (1) Number per second = $\frac{P\lambda}{hc}$ (1) Accept $P \div (\frac{hc}{\lambda})$ or equivalent		2		2	2	
		(ii)		Use of $p = \frac{h}{\lambda}$ seen in momentum change per sec = $N \times \frac{h}{\lambda}$ **ecf** (1) Simplified to $\frac{P}{c}$ **ecf** from (i) but not if left in terms of N (1) **First mark available even if no answer to (b)(i)**		2		2	2	
		(iii)		Force	1			1		
				Question 2 total	**5**	**4**	**4**	**13**	**7**	**0**